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MILITARY SPECIFICATION

AMINE BONDING AGENT

Reactivated after 30 April 2020 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the U.S. Army Combat Capabilities Development Command – Armaments Center (CCDC – AC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture and acceptance of two types of amine bonding agent (see 6.1).

Type I – For use in the M864 and XM1128 systems.
Type II – For use in the M913 and system.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army CCDC – AC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286 - Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E 168 - Standard Practices for General Techniques of Infrared Quantitative Analysis
ASTM E 300 - Standard Practice for Sampling Industrial Chemicals.
ASTM D 2196 - Standard Test Methods for Rheological Properties of Non-Newtonian Materials by Rotational Viscometer

(Copies of these documents are available online at https://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Composition. Type I amine bonding agent is the result of the reaction between tetraethylenepentamine and acrylonitrile to form Tepan. Type II amine bonding agent is the result of reaction of Tepan with glycidol to Tepanol. The Type I or Type II material shall conform to the applicable physical and chemical characteristics in Table I.

TABLE I. Chemical and physical requirements for amine bonding agents.

Property	Requirement				Applicable test paragraph
	Type I		Type II		
Total amine, eq/100g	1.1	1.6	0.8	1.2	4.5.2
Ammonia, ppm	-----0	500	-----	500	4.5.4
Moisture, %	-----	0.85	-----	0.5	4.5.5
Viscosity, 25˚C, cps	500	1,000	30,000	60,000	4.5.6
Hydroxyl content, eq/100g	-----	-----	0.4	0.6	4.5.9

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3.2 Infrared (IR) spectrum. The IR spectrum of the sample for Type I amine bonding agent shall contain the major peaks shown in Figure 1 when tested in accordance with 4.5.7.

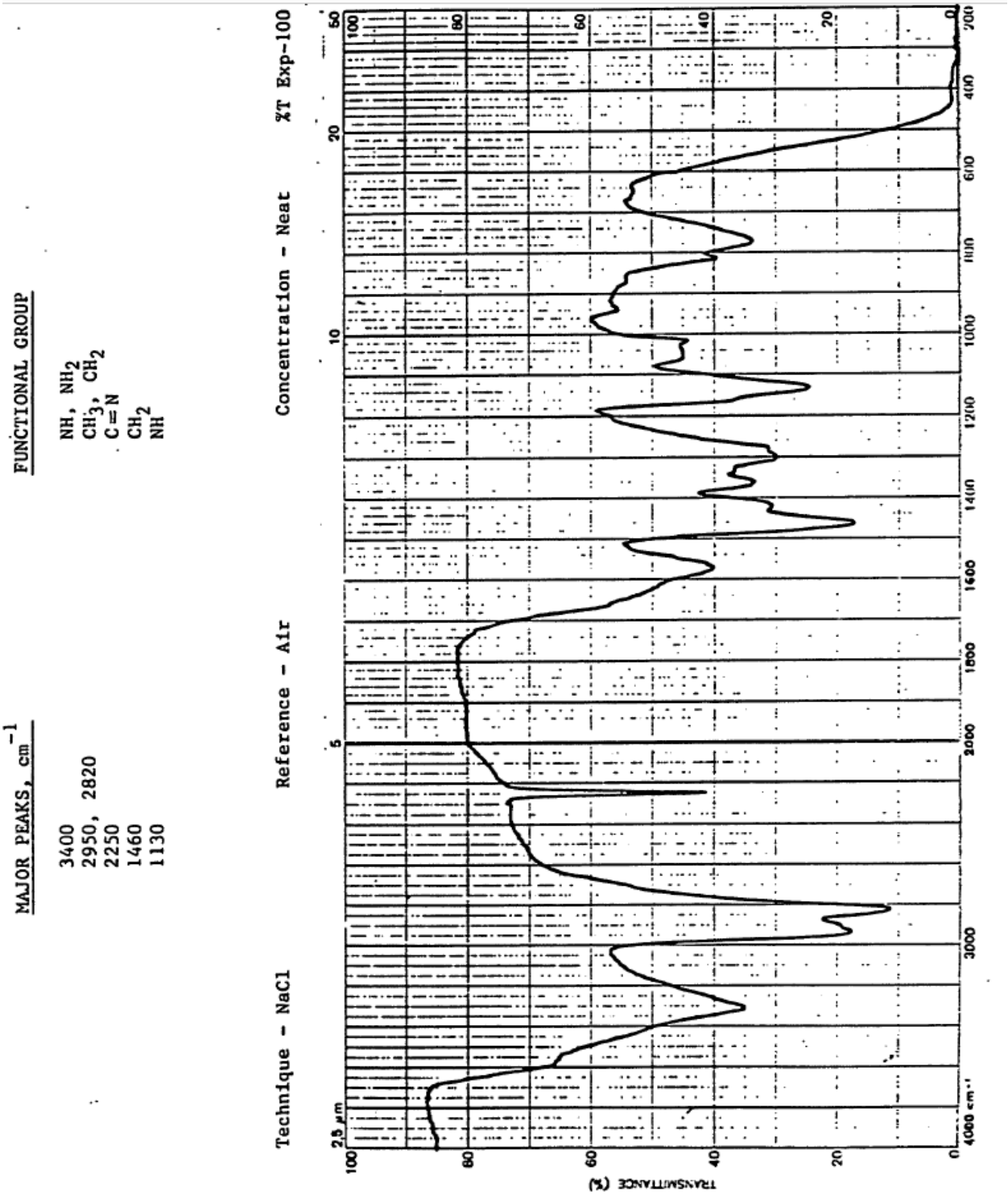


Figure 1. IR spectrum for Type I amine bonding agent.

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3.3 First article. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.4 Workmanship. All materials shall be uniform in quality and free from lumps, grit, visible impurities, foreign matter, or other defects that would render the material unsuitable for intended use.

4. VERIFICATION

4.1 DELETED.

4.1.1 DELETED.

4.2 Classification of inspections. The following inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of ten pounds of amine bonding agent. The sample shall be obtained from the first production lot which has been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to performed. The sample shall be subjected to any or all of the examinations and tests specified in Table II.

4.3.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected.

TABLE II. First article inspection.

Examination or test	Requirement paragraph	Inspection method paragraph
Total amine	3.1	4.5.2
Ammonia	3.1	4.5.4
Moisture	3.1	4.5.5
Viscosity	3.1	4.5.6
Infrared spectrum (Type I only)	3.2	4.5.7
Hydroxyl content (Type II only)	3.1	4.6.9

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4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of one or more batches of amine bonding agent produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of amine bonding agent that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. Lot formation shall be in accordance with MIL-STD-1916, Formation and identification of lots or batches.

4.4.2 Examination.

a. Workmanship. Examination for workmanship shall be conducted on each sample selected for testing in accordance with 4.4.3.1. If any sample fails to meet any requirement the batch represented by the sample shall be rejected.

4.4.3 Testing. PRECAUTION: This specification covers sampling and testing of toxic and hazardous material. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing the amine bonding agent.

4.4.3.1 Sampling. Approximately 500 grams of the composition shall be selected from each batch to be sampled using ASTM E 300 for liquids. Samples shall be selected for inspection in accordance with Table III. If any sample fails to meet any test requirement the batch represented shall be rejected. Each required test shall be performed in duplicate. No composite samples shall be used. All characteristics are defined as Major in accordance with MIL-STD-1916.

TABLE III. Classification of defects.

Number of containers in batch	Number of containers to be sampled
2 – 200	2
Over 200	5

4.4.4 Inspection equipment. For the performance of all tests and examinations specified in 4.4 and 4.5, commercial inspection equipment should be employed. The contractor shall have available, and utilize correctly, this equipment and is charged with the responsibility of assuring that proper calibration procedures are followed.

4.5 Test methods and procedures. All tests given in this section shall be performed using prescribed procedures for replicate determination given in standard analytical chemistry textbooks. Also, unless otherwise specified herein, all chemicals and reagents used in the tests shall be ACS Grade or Reagent Grade.

4.5.1 DELETED.

4.5.2 Total amine.

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4.5.2.1 Apparatus.

a. Beckman pH meter, or equivalent, equipped with glass-Ag/Ag/AgCl combination electrode or an automatic titrator;
b. Analytical balance, accurate to 0.0001 g
c. Cylinder, graduated, 100-mL
d. Bottles, 4-ounce or equivalent
e. Syringe, 1-mL, disposable

4.5.2.2 Reagents.

a. P-toluenesulfonic acid, 0.5N. Dissolve 95 g of p-toluenesulfonic acid monohydrate, AR grade, in approximately 900 mL glacial acetic acid. Slowly add 20 mL of acetic anhydride with constant stirring. Dilute to one liter with glacial acetic acid and mix.
b. Glacial acetic acid, AR grade.
c. Acetic anhydride, AR grade.

4.5.2.3 Standardization of p-toluenesulfonic acid solution.

a. Weigh, by difference, 0.9 ± 0.0001 g (W) of primary standard grade potassium hydrogen phthalate.
b. Dissolve in a 4-oz bottle using 50-75 mL of glacial acetic acid.
c. Place the sample to be titrated in a steel cup to reduce electrostatic interference.
d. Titrate potentiometrically with the p-toluenesulfonic acid solution using an automatic titrator.
    1. While stirring with a magnetic stirrer, add 1 mL p-toluenesulfonic acid solution to the bottle. Allow to stir and take the pH reading when it has stabilized. Record volume added and corresponding pH reading.
    2. Continue process, adding smaller aliquots of p-toluenesulfonic acid solution when nearing the equivalence point.
    3. Continue titrating until there is less than 0.2 change in pH upon addition of 1 mL p-toluenesulfonic acid solution.
    4. Graph total mL p-toluenesulfonic acid solution added versus pH.
e. Determine the volume required to reach the equivalence point by locating the midpoint of the break curve. Record the volume as V.
f. Calculate the normality of the p-toluenesulfonic acid solution as follows:

$$N = \frac{W}{Vx0.20423}$$

Where: W = weight of potassium hydrogen phthalate
V = volume of titrant used

g. Average the duplicate results
NOTE: p-toluenesulfonic acid should be restandardized every two weeks).

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4.5.2.4 Procedure.

a. Using a 1-mL syringe, weigh, by difference, 0.25 – 0.30 ± 0.0001 g (W) of sample into a 4-oz bottle.

b. Using a graduated cylinder, add 50 mL of glacial acetic acid and stir to dissolve.

c. Titrate the sample with the standardized 0.5N p-toluenesulfonic acid solution using the following procedure or an automatic titrator.

1. While stirring with a magnetic stirrer, add 1 mL of p-toluenesulfonic acid solution to the bottle. Allow to stir and take the pH reading when it has stabilized. Record volume added and corresponding pH reading.
2. Continue process, adding smaller aliquots of p-toluenesulfonic acid solution when nearing the equivalence point.
3. Continue titrating until there is less than a 0.2 change in pH upon addition of 1 mL p-toluenesulfonic acid solution.
4. Graph total mL p-toluenesulfonic acid solution added versus pH.

d. Determine the volume required to reach the potentiometric break point by locating the midpoint of the break curve.

Calculation:

$$\text{Total amine}, \frac{\text{eq}}{100\ \text{g}} = \frac{\text{VxN}}{\text{10xW}}$$

Where: V = volume of p-toluenesulfonic acid, mL
N = normality of p-toluenesulfonic acid
W = sample weight, g

4.5.3 DELETED.

4.5.4 Ammonia. Ammonia content shall be determined in accordance with the following:

4.5.4.1 Apparatus.

a. Ammonia gas-sensing electrode
HNU Model ISE-10-00, HNU Systems, Inc., 30 Ossipee Road, Newton, MA 02164
or
Orion Model 95-10, Orion Research, Inc., 380 Putnam Avenue, Cambridge, MA 02139
or

b. Meter, pH/mV, digital, Orion model 801A or equivalent
c. Stirrer, magnetic
d. Syringe, 3-mL, disposable
e. Cylinder, graduated, 100-mL
f. Balance, analytical, accurate to ±0.0001 g
g. Holder, electrode, 20° angle

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h. Flask, volumetric, 2000-, 1000-, 500-, and 100-mL

4.5.4.2 Reagents.

a. Sodium hydroxide, 10M. Add 40 g of AR grade NaOH to 80 mL of deionized (DI) water in a 100-mL volumetric flask, dissolve, and dilute to volume with DI water.
b. Standard ammonium chloride solution, 100- 50-, and 10-ppm as ammonia. To prepare a 100-ppm as NH_3 standard, add 0.6288 g of reagent grade ammonium chloride to about 1000 mL of DI water in a two-liter volumetric flask. Dissolve and dilute to volume with DI water. Dilute 500 mL of 100-ppm standard to one liter with DI water to prepare a 50-ppm as NH_3 standard. Dilute 100 mL of 100-ppm standard to one liter with DI water to prepare as 10-ppm as NH_3 standard.
c. Ammonium chloride, AR grade.
d. Sodium hydroxide, AR grade.
e. Standard laboratory glassware.

4.5.4.3 Calibration procedure.

NOTE: Recalibrate every two hours.
a. Assemble the ammonia electrode according to instruction manual.
b. Connect the electrode to the pH/mV meter, referring to instruction manual for specific directions.
c. Add 100±1 mL of the 10 ppm as NH_3 standard to a 150-mL beaker. With stirring, add 1-mL of 10M NaOH.
NOTE: Some magnetic stirrers generate sufficient heat to change solution temperature, which affects electrode response. To minimize this effect, place a piece of insulating material such as cork, polyurethane foam, or Styrofoam under the beaker.
d. Supply gentle agitation with the magnetic stirrer. After one minute, record the millivolt reading for the 10-ppm NH_3 standard. Rinse the electrode with DI water.
e. Repeat steps c and d for the 50-ppm and 100-ppm NH_3 standards.
f. Using a three cycle semi-logarithmic graph, construct a calibration curve.

4.5.4.4 Procedure.

a. Using a 3-mL syringe, weigh, by difference, 2-3 ± 0.001 g (W) of the sample into a 150-mL beaker.
b. Add 100±1 mL of DI water and stir to dissolve on thermally insulated magnetic stirrer.
c. Place the electrode in the sample solution.
d. Add 1 mL of 10M NaOH with stirring, and wait one minute for a stable reading. Record the display reading (R).
e. From the previously constructed calibration curve, interpolate, using R, to determine the sample solution concentration (C):

$$\text{Ammonia content, } NH_3 \text{ ppm} = \frac{C \times 100}{W}$$

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Where: C = sample solution of concentration, ppm
W = sample mass, g

4.5.5 Moisture. Determine moisture content in accordance with Method 101.5 of MIL-STD-286.

4.5.6 Viscosity. Determine viscosity in accordance with ASTM D 2196.

4.5.7 Infrared spectrum (Type I). Verify by examination that an infrared spectrum collected in accordance with ASTM E 168 matches a reference standard of amine bonding agent.

4.5.8 Workmanship. Visually examine the contents of the container sampled and the sample taken for the presence of foreign materials.

4.5.9 Hydroxyl content (Type II). The hydroxyl content for amine bonding agent shall be determined as follows:

4.5.9.1 Equipment.

a. Steam bath;
b. Flask, 250-mL with ground glass joints and stoppers
c. Buret, 50-mL
d. Pipette, 20-mL
e. Balance, analytical

4.5.9.2 Reagents.

a. Acetic anhydride, AR-grade
b. Pyridine, AR-grade
c. Sodium hydroxide, AR-grade
d. Acetylating reagent. Add 30 mL of acetic acid to 470 mL pyridine and mix thoroughly. Prepare fresh daily. Store in a dark bottle.
e. Sodium hydroxide, 0.5N in methanol, standardized
f. Phenolphthalein indicator, 1% in methanol
g. Deionized (DI) water

4.5.9.3 Procedure.

NOTE: Run sample in triplicate. Dry all glassware in 110˚C oven prior to use.

a. Weigh a 0.4 to 0.5 g sample of Type II amine bonding agent to the nearest 0.0001 g into a 250-mL flask. Record the weight as W_s.
b. Using a 20-mL volumetric pipette, add exactly 20 mL of acetylating reagent to each flask.
c. Prepare three blanks containing 20 mL aliquot of acetylating reagent and carry blanks through the entire treatment of the sample.
d. Moisten glass stoppers with pyridine and seat loosely.

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e. Assure sample is completely dissolved in acetylating reagent.
f. Heat samples and blanks gently on steam bath for one hour.
g. Add 50 mL of DI water, rinsing stopper and inside walls of flask.
h. Swirl the contents to ensure thorough mixing and heat on steam bath for five minutes.
i. Remove flasks and cool to room temperature.
j. Rinse stoppers and inside walls with 5-10 mL DI water.
k. Add one-half dropper of phenolphthalein indicator and titrate with standard 0.5N sodium hydroxide to the appearance of the first pink coloration. Record the volume of the blanks, V_b, and sampls, V_s.
NOTE: volume of the blanks must agree within 0.1 mL.
l. Calculate the hydroxyl content as follows:

$$\text{Hydroxyl content}, \frac{\text{eq}}{100\text{g}} = \frac{(V_b - V_s)xN}{10xW_s} - N_h$$

Where: V_b = the average volume of the titrant, sodium hydroxide, for the blanks
V_s = the average volume of the titrant, sodium hydroxide, for the sample
N = normality of the titrant, sodium hydroxide
W_s = weight of the sample
N_h = secondary amine content, eq/100g

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point' packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Type I amine bonding agent is intended for use in the base-bleed units of the M864 and XM1128; Type II amine bonding agent is intended for use in the M913 rocket motor.

6.2 Acquisition requirements. Acquisition documents should specify the following:
a. Title, number, and date of this specification.
b. Type of material required (see 1.1).

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c. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. Whether a first article sample is required (see 3.3).
e. Packaging requirements (see 5.1).

6.2.1 Data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, acceptance and description sheets prepared in accordance with MIL-STD-1171, or as directed in the contract should be supplied by the contractor.

6.3 Submission of inspection equipment designs for approval. Submit equipment designs, as required, to the Technical Agency (see 6.4)

6.4 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: Commander, CCDC – AC, ATTN: FCDD-ACE-QEE, Picatinny Arsenal, NJ 07806-5000.

6.5 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to the Technical Agency (see 6.4). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.6 Subject term (key word) listing.

Igniter
Base bleed
Propellant
Burn rate modifier

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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